

System Architecture

- **Input State (6D):** batch size, wait time, queue length, time since last skip, arrival rate, system load
- **Predictive Layer:** grouped linear predictor estimates near-future demand
- **Decision Layer:** Predictive Dyna-Q policy chooses WAIT (0) or SKIP (1)
- **Environment Feedback:** reward and next state update Q-table + planning model

Methodology

- Train/evaluate on poisson, bursty, and time_varying traffic
- Evaluate across 5 seeds with aggregated metrics
- Core metrics: mean reward, avg wait, P95 wait, avg batch size, throughput, SLO violation rate

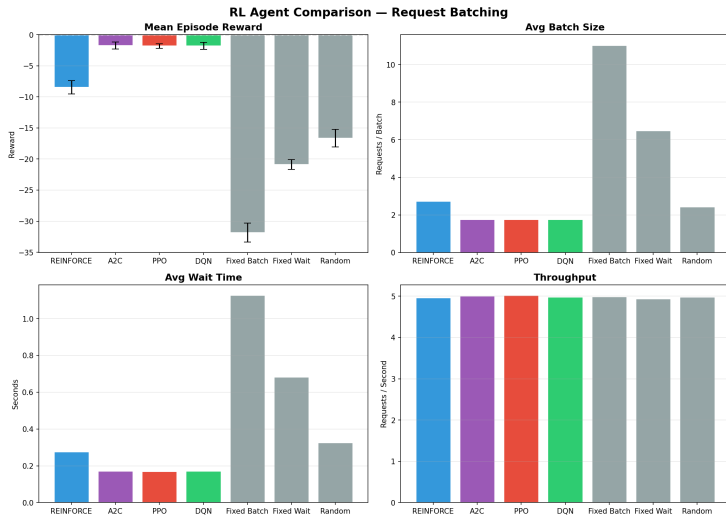
Scores and Final Metrics (Aggregated)

Model	Mean Reward	Avg Wait (s)	P95 Wait (s)	SLO Viol. (%)
REINFORCE	-2.07	0.244	0.453	0.0
A2C	-25.81	1.099	2.018	54.9
PPO	-26.22	1.109	2.025	55.5
Predictive Dyna-Q	-29.96	0.587	1.518	24.5
Fixed Wait (1.0s)	-14.70	0.646	1.100	22.1
Random	-10.82	0.330	0.869	3.2

Notes:

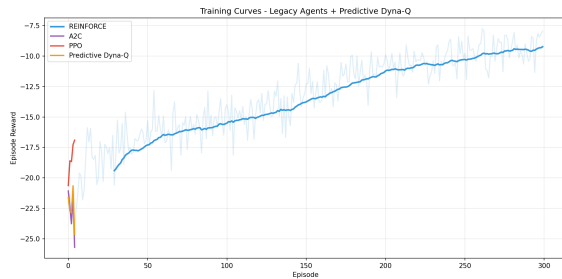
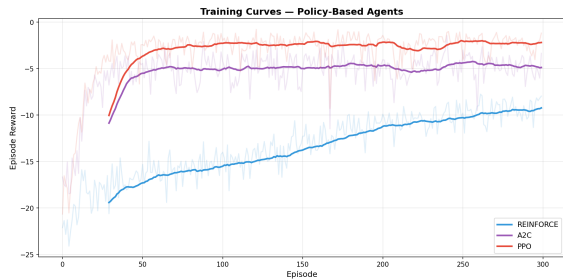
- Throughput for all methods remains around 6.6–6.8 req/s.
- Main optimization target was latency-quality tradeoff under variable demand.

Agent comparison plot



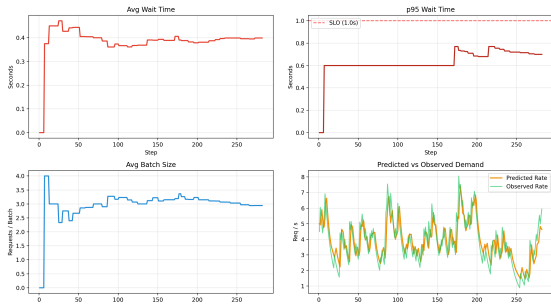
comparative metrics plot

Training curves



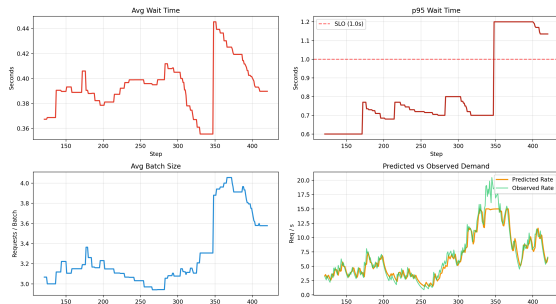
Live simulation plots

LIVE SIMULATION: Predictive Dyna-Q
Reqs Served: 103 | Batches Sent: 35 | Throughput: 3.7 r/s | SLO Violations: 0.0% | Action: WAIT



103 requests served.

LIVE SIMULATION: Predictive Dyna-Q
Reqs Served: 254 | Batches Sent: 71 | Throughput: 6.0 r/s | SLO Violations: 6.3% | Action: WAIT



254 requests served.